# RESEARCH STATEMENT

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## 1. Research Interests and Vision

**Research Interests:** As an AI research scientist, I am passionate about advancing agent-based task automation toward human–agent collaboration and exploring new frontiers of perception in autonomous agents operating in complex, real-world environments through multimodal large language models. As foundation models move from answering queries to executing multi-step workflows, the central research problem becomes making their autonomy reliable, verifiable, and deployable — above all on the temporal, numerical, and multimodal data that real-world decisions depend on, where errors are costly and verification is hard. Stated in one line, my research builds self-improving AI agents that reason reliably over temporal, multimodal, and physical-world data.

**Research Vision:** Agentic AI is advancing along three directions — greater autonomy, broader real-world perception, and physical-world deployment — and my long-term plan follows all three.

- **Greater autonomy.** AI systems are shifting from executing narrowly specified tasks to planning their own work, building their own tools, anticipating their own success, and deciding what to work on. Realizing this requires multi-agent systems with explicit planning, verification, and self-assessment.
- **Broader real-world perception.** The data that matters in practice is overwhelmingly temporal, numerical, and multimodal — exactly where foundation models remain brittle. New representations, architectures, and verification mechanisms are needed so that agents can reason over such data and know when their conclusions hold.
- **Physical-world deployment.** The next generation of agents will run on resource-constrained devices, continuously and proactively, embedded in the physical environment rather than confined to a chat session — demanding efficient on-device models, latency-bounded reasoning, and evaluation science for always-on agents.

**In brief:** to date I have built foundations for time-series representation and neural architecture search, multi-agent systems that automate the machine-learning lifecycle and anticipate their own performance, and spatiotemporal models for real-world urban data. Going forward, I will build agents that improve through verified tool use, reason from evidence rather than consensus, and deploy safely on resource-constrained devices.

## 2. Research Accomplishments

My work joins two traditions that rarely meet: the statistical foundations of time-series data mining and the systems engineering of autonomous agents. During my Ph.D. at KAIST's Data Mining Lab, advised by Prof. Jae-Gil Lee, I developed methods for spatiotemporal data mining and for automating neural-architecture design over time series; as a research scientist at DeepAuto.ai, I extended this foundation to multi-agent LLM systems that automate the AI/ML lifecycle. Few researchers work across both traditions, and that vantage point is what lets me study a central problem in agentic AI: how to build agents that are not merely capable, but reliable on the temporal and multimodal data that real decisions depend on.

My contributions span three areas — AI agents for task automation, time-series analysis and multimodal reasoning, and spatiotemporal data mining. I foreground the flagship result of each below; related projects appear as supporting work and in the publication list.

## 2.1 AI Agents for Task Automation

Building machine-learning solutions remains gated behind scarce expertise and fragmented across many manual steps. My first area asks whether LLM agents can own this lifecycle end to end — planning the work, building the tools to do it, anticipating their own success, and deciding which problems to solve.

### 2.1.1 Full-Pipeline AutoML with Multi-Agent LLMs

Existing AutoML systems target a single stage of the pipeline and require expertise to configure. I led the design of **AutoML-Agent** [1], a multi-agent LLM framework for the full AutoML pipeline, from data retrieval to deployment-ready models: it uses retrieval-augmented planning to explore competing strategies rather than committing to one, decomposes each into sub-tasks solved in parallel by specialized agents, and applies multi-stage verification to guide code generation. Across seven downstream tasks and fourteen datasets it attains a higher end-to-end success rate than prior single-stage methods, extending my earlier **Conversational AutoML** [14], which first framed model construction as a natural-language dialogue.

### 2.1.2 Performance Prediction for Agentic Workflows

Optimizing an agentic system is costly because of the vast space of configurations, prompts, and communication patterns. I developed **Multi-View Encoders** [2], a lightweight predictor that encodes a candidate workflow from three complementary views — code architecture, textual prompts, and the agent interaction graph — and, through cross-domain unsupervised pretraining, selects high-performing configurations under scarce labels without expensive trial-and-error (Outstanding Paper, ICML 2025 Workshop on Multi-Agent Systems).

### 2.1.3 Toward Greater Autonomy

Three further projects push agents from executing tasks to owning them. **MONAQ** [3] recasts on-device architecture search as multi-objective *querying* driven by LLM agents over numerical, visual, and textual views of a model, discovering networks that beat both handcrafted and NAS baselines while staying efficient on constrained hardware. **Tooler-Agents** (ongoing) builds a persistent, dependency-aware toolset on demand, admitting only tools that compose under sandboxed tests and keeping per-query planning cost independent of toolset size. And **Proactive Problem Solver** (ongoing) formalizes *ambient proactivity* — an always-on system that decides *what* to investigate under policy-gated autonomy, evaluated by 24-month archival replay with an explicit leakage audit. Together they sketch an agent that plans its work, builds its own tools, predicts its own success, and chooses its own problems.

## 2.2 Time-Series Analysis and Multimodal Reasoning

An agent is only as reliable as its reasoning over real-world data, which is overwhelmingly temporal, numerical, and multimodal. My second area builds the representation, architecture, and reasoning substrate for such data, with the benchmarks and evaluation rigor needed to know when it works.

#### 2.2.1 Foundations of Representation and Architecture

To organize a fast-growing field, I led the first survey of universal time-series representation learning [4], whose taxonomy of 127 methods turns on three design elements — the training data, the network architecture, and the learning objective (*ACM Computing Surveys*, impact factor 28.0). I then made the architecture itself a learning problem: **PASTA** [6] searches temporal-connectivity spaces for anomaly detection, improving F1 by at least 13.6% over the runner-up; **TFAS** [5] is a unified zero-shot NAS whose time-frequency scoring improves five task types by up to 23.6%; and **Time-PEFT** [7], with colleagues, uses data-driven complexity to guide parameter-efficient fine-tuning of time-series foundation models, improving forecasting by up to 2.51×.

#### 2.2.2 Multimodal Reasoning, and Its Limits

My current focus is multimodal reasoning over time series. **TS-Debate** [11] (ongoing) formalizes it as evidence arbitration: modality-specialized agents generate evidence in isolation, and a Verification–Conflict–Calibration protocol resolves their claims by code execution and numerical lookup rather than persuasion or voting, improving accuracy by 25.65% on average over the strongest baseline across twenty tasks. Crucially, the companion **More Modalities, More Problems** (ongoing) is a rigorous negative result: naively scaling debate across modalities can cost roughly 22× the tokens and 12× the time of a strong single-agent baseline, with 74% of errors from a shared modality bias that prompts do not remove. Two further projects add theory and evaluation — **Visual Anchoring** derives a closed-form Lipschitz bound on how a scenario's influence decays over a forecast horizon and turns it into a chart-anchored prompt, and **MacroLens** contributes a leakage-controlled benchmark of 4,416 U.S. equities with build-time invariants against lookahead. Publishing the boundaries of a method, not only its wins, is central to how I work.

### 2.3 Spatiotemporal Data Mining

My research is rooted in mining the traces real-world activity leaves behind — heterogeneous, physically-grounded, privacy-sensitive signals at city scale — and this grounding shapes my agenda for agents that act in the physical world. **DF-TAR** [9] fuses dangerous-driving telemetry with environmental features for citywide traffic-accident risk prediction, improving baseline accuracy by up to 54%, and **MG-TAR** [8] extends it with multi-view graph convolutional networks over heterogeneous urban data; with collaborators, **ZeroPOIRec** [10] shows that pretrained LLMs can serve as zero-shot point-of-interest recommenders over individual mobility histories, outperforming state-of-the-art methods while requiring no training on shared visit data.

## 3. Future Research

My future research unifies these areas into one program — self-improving agents that reason reliably over real-world data — pursued along three thrusts, the first of which — self-improving systems — is the heart of the agenda, with the other two making its reasoning reliable and its deployment safe. For each I give the defining question, the systems I will build, how I will evaluate success with reliability as a first-class outcome, a near-term deliverable, and why it matters.

### 3.1 Self-Improving Agentic Systems

**Question.** Can agents build, verify, and reuse their own tools and workflows so that competence compounds across tasks, instead of restarting each task from scratch? **Approach.** Building on AutoML-Agent [1], Tooler-Agents, and Multi-View performance prediction [2], I will develop verified tool synthesis, workflow predictors that estimate utility before paying to run a candidate, and memory that retains and retrieves what works. **Evaluation.** End-to-end task success, cost, and reliability under distribution shift, and whether performance compounds over a stream of tasks rather than plateauing. **Near-term.** An open-source self-improving AutoML agent and a public benchmark for agentic-workflow performance prediction. **Impact.** This turns today's one-shot pipelines into systems that measurably improve with use.

### 3.2 Reasoning Grounded in Verification, Not Persuasion

**Question.** My negative result in *More Modalities, More Problems* shows that multi-agent debate can amplify a shared bias rather than correct it; how do we make agent reasoning settle on evidence instead of consensus? **Approach.** I will design reasoning architectures whose conclusions are adjudicated by programmatic verification, with modality-specific uncertainty, calibration, and theory-backed control — extending the Verification–Conflict–Calibration protocol of TS-Debate [11] and the drift bounds of Visual Anchoring. **Evaluation.** Beyond accuracy, I will measure calibration and auditability — whether each conclusion is traceable to a verifiable check — and robustness when modalities conflict. **Near-term.** An open-source verification-grounded reasoning library and an auditability-and-calibration evaluation suite for multimodal time-series reasoning. **Impact.** Reasoning that is auditable rather than merely persuasive is a precondition for deploying agents in high-stakes scientific, financial, and operational decisions.

### 3.3 Proactive, On-Device, Ambient Agency

**Question.** How can agents run continuously in the physical world under latency, memory, energy, safety, and privacy constraints? **Approach.** Pursuing the **Agent of Things** [15] agenda, I will build hardware-aware architecture search that extends MONAQ [3], tiered and latency-bounded reasoning for duty-cycled hardware, open protocols that let heterogeneous agents coordinate across vendors, and reversibility mechanisms that make autonomous action safe to undo. **Evaluation.** Safety benchmarks, reversibility guarantees, and scaling laws for on-device inference, developed on the same timeline as the capability science. **Near-term.** An on-device agent prototype and a safety-and-reversibility benchmark for always-on physical agents. **Impact.** This connects agentic AI to deployment at the scale of the more than eighteen billion connected devices already in the world.

Across these thrusts my goal is a single one: agents that earn their autonomy — capable enough to act on the temporal, multimodal, and physical data that real decisions depend on, and rigorous enough to know when they are right.

## References


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